

Brent Pappas

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EDUCATION

PhD. Computer Science. GPA: 3.983. University of Central Florida. Orlando, Florida. August 2021 - Present.

Master of Science. Computer Science. GPA: 3.983. University of Central Florida. Orlando, Florida. August 2021 - December 2025.

Bachelor of Science. Computer Science. GPA: 3.849. University of Central Florida. Orlando, Florida. August 2017 - May 2021.

TEACHING

Spring 2026	Instructor of Record. COP-3402 Systems Software. UCF. Course homepage . Lecture Recordings . Student perception of instruction, quantitative results . Student perception of instruction, qualitative results .
Fall 2025	Graduate Teaching Assistant. COP-3402 Systems Software. UCF. Student perception of instruction
Spring 2025	Graduate Teaching Assistant. COP-3402 Systems Software. UCF.
Spring 2021	Peer Tutor Mentor. COP-3502 Computer Science 1 & COP-3330 Object Oriented Programming. UCF.
Spring & Fall 2020	Peer Tutor. COP-3502 Computer Science 1 & COP-3330 Object Oriented Programming. UCF.
Fall 2019	Peer Tutor. COP-3502 Computer Science 1. UCF.

PUBLICATIONS

- 2026 **Build Code is Still Code: Finding the Antidote for Pipeline Poisoning**: by Brent Pappas and Paul Gazzillo. ICSE NIER. Acceptance rate of 33.11% (50 / 151). [Pre-print](#). [Artifact](#). Badges: Available and reusable. [Presentation](#). [Slides](#).
- 2024 **Semantic Analysis of Macro Usage for Portability**: by Brent Pappas and Paul Gazzillo. ICSE. Acceptance rate of 22.26% (234 / 1051). [Pre-print](#). [Artifact](#). Badges: Available and reusable. [Presentation](#). [Slides](#).
- 2023 **Holy Macroni! A recipe for progressive language enhancement**: by Brent Pappas. Blog post for Trail of Bits. [Blog post](#). [Artifact](#).
- 2022 **SugarC: Scalable Desugaring of Real-World Preprocessor Usage into Pure C**: by Zach Patterson, Zenong Zhang, Brent Pappas, Shiyi Wei, and Paul Gazzillo. ICSE. Acceptance rate of 28.51% (197 / 691). [Pre-print](#). [Artifact](#). Badges: Available and reusable. [Presentation](#).

SERVICE

- March 2026 **NAC - NAPC Volunteer.** Orlando, Florida.
 - **Inflating Balloons:** Inflated competition balloons for the 2026 North American Championship and Programming Camp (NAC-NAPC), hosted at UCF.
- October 2025 **UCF CS Grad Symposium 2025 Student Panelist.** Orlando, Florida.
 - **Early PhD Advice:** Recommended that new graduate students remain humble and start honing their communication skills immediately after starting a PhD.
- August 2025 **UCF GradLaunch PhD Panelist.** Orlando, Florida.
 - **Advising:** Instructed new graduate students how to manage their time and organize their day in order to be productive researchers.
- August 2025 **UCF "Surviving Graduate School" Panelist.** Orlando, Florida.
 - **Community engagement:** Taught prospective graduate students how to become active in the UCF community.
- June 2025 **Camp Connect Lab Leader.** Orlando, Florida.
 - **Human Computer Interaction Lessons:** Gave primary school students hands-on training with biometric equipment, including respiration belts, EDA and ECG sensors, fNIRS headbands, and eye-tracking sensors.

May 2025	UCF Graduate School Info Session PhD Panelist. Orlando, Florida. • Advising: Answered undergraduate students' questions about UCF's Computer Science PhD program.
April 2024	UCF CECS Senior Design Judge. Orlando, Florida. • Judging: Assessed capstone projects for seniors of the College of Engineering and Computer Science.
October 2022	UCF CECS Career Kickoff Alumni Mentor. Virtual. • Mentoring: Advised UCF freshmen on goal-setting, resume-building, and overall professionalism.
May 2024	NAC - NAPC Volunteer. Orlando, Florida. • Competition setup: Moved computers and cables to the competition room for the 2024 North American Championship and Programming Camp (NAC-NAPC), hosted at UCF.
2024	OOPLSA 2024 Artifact Evaluation Committee Member. Virtual. • Artifact Review: Reviewed code associated with papers accepted to OOPLSA 2024. Distinguished reviewer.
June 2023	Camp Connect Lab Leader. Orlando, Florida. • Javascript lessons: Taught primary school students the basics of computer programming in JavaScript.
Nov - Dec 2022	EuroSys 2023 Shadow Program Committee Member. Virtual. • Paper review: Reviewed submissions to EuroSys 2023.
June 2021	Camp Connect Lab Leader. Orlando, Florida. • Computer programming lessons: Engaged primary school students in computer programming lessons through a series of programming-themes games.

PRESENTATIONS

2026	Build Code is Still Code: Finding the Antidote for Pipeline Poisoning. Presentation at the 2026 International Conference on Software Engineering in Rio de Janeiro, Brazil. YouTube recording . Slides .
2026	Curing Pipeline Poisoning. 3 minute thesis on my research on securing build systems against pipeline poisoning with build phase isolation. Presented at the CRA 2026 Grad Cohort Workshop. Recording link .
2026	Invited Talk - The PhD Life. Guest lecture on graduate student life presented to Sonoma State University's 2026 Spring Colloquium, hosted by Dr. John Sullins. Slides .
2025	Invited Talk - The PhD Life. Guest lecture on graduate student life presented to Dr. Suzanne Rivoire's undergraduate course, Computing Professions, at Sonoma State University. Slides .
2024	Semantic Analysis of Macro Usage for Portability (ICSE 2024). Presentation at the 2024 International Conference on Software Engineering in Lisbon, Portugal. Recording .
2023	Semantic Analysis of Macro Usage for Portability (CAE-R) Virtual presentation for Center of Academic Excellence in Cyber Research (CAE-R). Recording ; jump to 25:38 .

POSTERS

2024	Holy Macroni! Finding Macro Usage Errors in the Linux Kernel. 2024 International Symposium on Secure and Private Execution Environment Design. Link .
2022	Cpp2C: Transforming C Preprocessor Macros to C Language Constructs. 2022 University of Central Florida Student Scholar Symposium. Link .

HONORS AND AWARDS

2026	GTA Excellence Award (University level). For providing exceptional teaching assistance to the course Systems Software. Awarded to 1 out of 800+ graduate teaching assistants. Award letter .
2026	GTA Excellence Award (College level). For providing exceptional teaching assistance to the course Systems Software. Award letter .
2026	CRA-WP 2026 Grad Cohort Workshop Invitee. The CRA-WP arranged and paid for me to travel to the 2026 CRA-WP Grad Cohort Workshop in Seattle, Washington, because of my superior workshop application.
2024	UCF 3MT Finalist. For effectively communicating my research to a general audience in a 3 minute talk. Video . Certificate .
2024	Distinguished Artifact Reviewer. For my attention to detail while reviewing artifacts for OOPLSA 2024. Link .

2024	FCI Student Scholarship. For my interest in interdisciplinary research, scholarship, and collaboration. \$5000 award.
2021	Cum laude honors. For having a cumulative undergraduate GPA greater than 3.8.
2017 - 2021	Deans' list. For continued undergraduate scholastic achievement.
2017	Burnett Honors Scholar. For academic talent, motivation, intellectual curiosity and creativity.
2017	Provost Scholarship. For outstanding academic performance in high school. Completely covered my undergraduate tuition fees at the University of Central Florida.

CERTIFICATIONS

Completed higher education training, [Preparing Tomorrow's Faculty](#)

Level 1 College Readiness and Learning Assessment (CRLA) certified tutor.

Level 2 College Readiness and Learning Assessment (CRLA) certified tutor.

UCF Student Enhancement Training (SET) program certified.

WORKSHOPS

March 2026	2026 CRA-WP Grad Cohort Workshop. Seattle, Washington.
Oct - Nov 2025	2025 CRA-WP Virtual Mentoring Series Workshops. Virtual.
May 2025	2025 CRA-E Career Landscape Workshop. Virtual.
June 2022	Programming Language Mentoring Workshop (PLMW). San Diego, California.

COMPLETE WORK EXPERIENCE

Jan 2026 - May 2026	Graduate Teaching Associate (Instructor of Record) for COP-3402 Systems Software University of Central Florida. Orlando, Florida. • Lecturing. Twice a week, lectured to a class of 250 students on systems software topics, including Linux command line tools, C system calls, git, make, and compiler design with ANTLR. • Exam Creation. Constructed and graded midterm and final exams. • Office hours. Hosted four 45-minute long office hour sessions each week to resolve student issues with assignments and provide supplemental instruction on topics such as how to edit software with Vim.
Aug 2023 - Apr 2024	Graduate Teaching Assistant for COP-3402 Systems Software University of Central Florida. Orlando, Florida. • Teaching. Led weekly labs on the Linux command line, C system calls, git, make, and compiler design. • Exam Proctoring. Answered student questions during exams and monitored students to prevent cheating. • Student perception of Instruction. Link.
May 2023 - Aug 2023	Software Engineering Intern Trail of Bits. Remote. • C Language Toolsmithing. Created Macroni, a C static analysis tool that combines Trail of Bits' tools PASTA and VAST to lower C preprocessor macros down to MLIR. Macroni is available on GitHub. • Peer-reviewed Development. Leveraged GitHub to push changes to company repos, submit code for code review, and finally merge changes into mainline branches. • Research Discussion. Exhibited research results and Macroni to team members and company CEO, Dan Guido, for advice and criticism.
Jul 2021 - Jul 2023, May 2025	Graduate Research Assistant University of Central Florida. Orlando, Florida. • Software Engineering Research. Stay up-to-date on the latest academic developments in software engineering. • Software Engineering. Implement research tools in C++ and Python. • Conduct Experiments. Write Python and Bash scripts to automate running experiments and analyzing data.

- Sep 2019 - Jul 2021** **Quality Assurance Coordinator**
 SP+. Orlando, Florida.
- **Agile Software Development.** Engaged in agile software development methods within a small team to QA test software before moving on to later phases of development.
 - **Testing Automation.** Refactored and wrote Python scripts to automate testing of various online user applications.
 - **Mobile Application Development.** Tested company Android apps for compatibility with various versions of Android.
- Jan 2021 - May 2021** **Peer Tutor Mentor**
 Student Academic Resource Center at the University of Central Florida. Orlando, Florida.
- **Tutor Mentoring** Observed other tutors' sessions and provided feedback as to how they may improve their communication and tutoring skills.
- Aug 2018 - May 2021** **Peer Tutor**
 Student Academic Resource Center at the University of Central Florida. Orlando, Florida.
- **Group tutoring.** Utilized Socratic teaching methods to teach Computer Science 1 and Object Oriented Programming to groups of 5 to 20 students at a time.
 - **Faculty Collaboration.** Collaborated with UCF teaching staff to enhance tutoring session quality.

ADVISING

UNDERGRADUATE RESEARCHERS

- Aaron Burgess.** 2026 - Present. Human factors in software engineering.
- Thiago Airoidi.** 2026 - Present. Analysis and security of software development pipelines.
- Rasmin Ahmed.** 2026 - Present. Analysis and security of software development pipelines.
- Daniel Henriquez.** 2026 - Present. Analysis and security of software development pipelines.
- Brayden Coggin.** 2026 - Present. Analysis and security of software development pipelines.
- Thai Nguyen.** 2025 - 2026. Human factors in software engineering and analysis and security of software development pipelines.
- Nikola Krstajic.** 2025. Human factors in software engineering.
- Michael Johnson.** 2025 - Present. Human factors in software engineering and analysis and security of software development pipelines.
- Layla Le.** 2025. Human factors in software engineering.
- Kenneth Valladares.** 2025. Human factors in software engineering.
- Julian Velez.** 2025. Human factors in software engineering.
- Jorge Toledo.** 2025. Human factors in software engineering.
- Hasini Daliboyina.** 2025. Human factors in software engineering.
- David Gusmao.** 2025 - Present. Human factors in software engineering and analysis and security of software development pipelines.
- Anju Thomas.** 2025. Human factors in software engineering.
- Adam Betinsky.** 2025 - Present. Human factors in software engineering and analysis and security of software development pipelines.
- Zachary Burkett.** 2024. Semantics-aware translation of C preprocessor.
- Joseph Zalusky.** 2024. Semantics-aware translation of C preprocessor macros.

PROJECTS

- 2026 Foreman.** University of Central Florida. A Python tool that enables developer to specify the commands file access permissions of phase of a project's build system, to protect these phases against pipeline poisoning attacks. Foreman is the tool associated with 2026 ICSE NIER paper, *Build Code is Still Code: Finding the Antidote for Pipeline Poisoning*.
- 2024 Macroni.** Trail of Bits. A C static analysis tool that combines Trail of Bits' tools PASTA and VAST to lower C preprocessor macros down to MLIR. [GitHub](#).
- 2024 Maki.** University of Central Florida. A C++ Clang plugin that analyzes hows C preprocessor macros affect the C AST. Maki provides language porting tools information they can use to port macros to target languages in a way that preserves not just macro behavior, but also macro abstractions. Maki is the tool associated with 2024 ICSE paper, *Semantic Analysis of Macro Usage for Portability*, for which I am the lead-author. [GitHub](#).

SKILLS SUMMARY

Languages: Python3, C/C++, Bash, and HTML/CSS/JavaScript

Tools: Git, GitHub, and LLVM/Clang

Platforms: Linux, Web, and Windows

Soft Skills: Leadership, Writing, Communication, and Time management